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I hereby declare that I have written this thesis independently and that I have not previously submitted it as an examination performance at any other university or in any other study programme.

I confirm that I have fully and correctly cited all sources and aids used in the preparation of this thesis.

All passages in the thesis that are quoted or paraphrased from publications or other external sources are clearly identified as such.

Use of AI-Supported Tools:

I declare that I have not used any AI-supported tools, or only the following AI-supported tools, in the preparation of this thesis:

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- ☒ I used the following AI-supported tools: **Google Gemini, OpenAI ChatGPT**

The AI-supported tools were used exclusively for the following purposes:

Assistance with grammar check, restructuring sections for clarity, and proofreading. The scientific ideas, research design, implementations, data analyses, interpretations, and conclusions are entirely my own.

All content generated by AI has been checked for accuracy and authorship and labelled accordingly. I have read and understood the instructions of BTU Cottbus-Senftenberg regarding the use of generative AI tools in scientific work. (Source: <https://www.b-tu.de/forschung/profil/forschungsethik/umgang-mit-ki-tools/#c369137>)

I have been informed that the Chair of Wireless Systems reserves the right to check the thesis for plagiarism and the use of AI-based tools. Should this examination reach such a conclusion, I will be given the opportunity to justify myself in a detailed interview and through self-disclosure.

Cottbus, October 8, 2026
Place, Date

Signature of the Author (Hymavathy Vanam)